SERVICENOW BASED LICENSED SOFTWARE INSTALLATION REQUEST APPLICATION

Project Report

Submitted by

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Service Catalog Development | Catalog Item Configuration | Flow Designer | Approval & Fulfillment

Project Title

An employee requests installation of licensed software through the Service Catalog

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1. INTRODUCTION

The Licensed Software Installation Request Management System is a ServiceNow-based solution designed to streamline the process of requesting, approving, and fulfilling licensed software installations within an organization. The system provides employees with a centralized Service Catalog item through which they can request approved software in a structured and standardized manner.

This project demonstrates the complete life cycle of a licensed software request, beginning with catalog item submission and continuing through data validation, license-related approval, fulfillment assignment, status tracking, and closure. The solution uses ServiceNow capabilities such as Service Catalog, catalog variables, approvals, Flow Designer, notifications, task fulfillment, and Update Set management.

Project Overview

Item	Description
Platform	ServiceNow
Primary Process	Licensed Software Installation Request Management
Service Channel	Service Catalog / Employee Self-Service
Scenario	Employee requests installation of licensed software for official work
Primary Roles	Employee, Manager/Approver, IT Software Support Team, ServiceNow Administrator
Main Activities	Catalog item creation, variable configuration, validation, approval, fulfillment, testing and deployment preparation
Key Records	Catalog Item, Requested Item (RITM), Approval, Catalog Task, User Record, Software/License information
Outcome	A structured software installation request process with controlled approval and fulfillment
Overall Benefit	Reduced manual effort, improved license governance, traceability and faster software provisioning

2. PROJECT OBJECTIVES AND SCOPE

2.1 Business Objectives

- Provide employees with a centralized Service Catalog option for requesting licensed software.
- Standardize software request information through controlled catalog variables and mandatory fields.
- Ensure that software installation requests are reviewed before fulfillment.
- Support license governance by capturing software name, version, business justification and user details.
- Reduce manual email-based requests and improve visibility of request status.
- Create a traceable request record from submission through installation and closure.

2.2 Functional Scope

- Service Catalog category and catalog item creation.
- Catalog variables for employee, software name, version, installation type, business justification and required date.
- Mandatory field and data validation configuration.
- Approval flow for manager or authorized software approver.
- Automatic creation of fulfillment tasks for the IT/software support team.
- Request status tracking using Requested Item and Catalog Task records.
- Notifications for submission, approval, rejection and completion.
- Update Set management for moving project configuration between instances.

2.3 Stakeholders

Stakeholder	Role	Needs / Expectations	Automation Impact
Employees	Submit software requests	Easy request submission and status visibility	Faster and standardized requests
Manager / Approver	Review business need	Clear request information and approval action	Reduced manual coordination
IT Software Support	Install and fulfill software	Accurate software and user information	Streamlined fulfillment
Software / License Admin	Validate license availability	License-aware request information	Better compliance and governance
ServiceNow Administrator	Maintain configuration	Secure and maintainable application	Centralized administration
Organization / Management	Monitor software usage and requests	Reporting and compliance visibility	Better planning and governance

3. REQUIREMENT ANALYSIS AND PLANNING

The requirement analysis phase defines the business need, request information, approval requirements, fulfillment responsibilities and implementation roadmap for the Licensed Software Installation Request Management System. Proper planning ensures that the Service Catalog request is standardized and that every request follows a controlled path from employee submission to fulfillment.

3.1 Execution Roadmap

- Requirement Analysis and Project Initialization
- Service Catalog Category Creation
- Licensed Software Catalog Item Creation
- Catalog Variable and Data Configuration
- Form Validation and Mandatory Field Configuration
- Approval Flow Design and Implementation
- Fulfillment Task Creation and Assignment
- Notification Configuration
- Testing and Validation of Request Creation
- Update Set Verification and Configuration Capture
- End-to-End Testing and Deployment Validation

3.2 Software Installation Request Scenario

An employee requires licensed software for official work and submits a request through the Service Catalog. The employee selects the required software, provides the business justification, confirms the installation target and submits the request. ServiceNow creates the Requested Item and routes it through the configured approval process. After approval, a fulfillment task is assigned to the IT software support team. The team validates license availability and performs the installation. The request is then updated and closed after successful fulfillment.

3.3 Request Lifecycle

The expected lifecycle begins when an employee opens the Service Catalog and selects the Licensed Software Installation item. The employee enters the required request information and submits the form. ServiceNow validates the information and creates the request. Approval is obtained from the configured approver. If approved, fulfillment is assigned to the IT software support team. If rejected, the requester is notified and the request is closed or cancelled. After installation, the fulfillment task and Requested Item are updated to reflect completion.

Figure 1 : Licensed Software Request Lifecycle Employee → Service Catalog → Validation → Approval → IT Fulfillment → Installation → Closure

4. SERVICE CATALOG CONFIGURATION AND DATA MODEL

Service Catalog configuration provides the foundation for presenting the licensed software request as an employee self-service option. The catalog item collects the information required by approvers and IT fulfillment teams while ServiceNow maintains the related Requested Item and Catalog Task records.

4.1 Catalog Configuration

- Open ServiceNow Service Catalog configuration or Service Catalog in the appropriate application scope.
- Create or select a suitable catalog category such as Software / IT Services.
- Create a catalog item named Licensed Software Installation Request.
- Provide a clear short description explaining that employees can request approved licensed software.
- Configure item availability for the intended employee audience.
- Configure the catalog item variables, mandatory fields, and default values.
- Save and verify that the item is visible in the Service Catalog.

4.2 Catalog Item and Data Model

The catalog item uses variables to collect structured request information. The employee submission results in a Requested Item (RITM), which can be processed through approvals and fulfillment tasks. The design separates the employee-facing request form from the fulfillment work performed by the IT team.

Variable / Record	Purpose	Example
Requested For	Identifies the employee receiving the software	Selected User
Software Name	Identifies the requested licensed application	Approved Software
Software Version	Captures required version	Current / Approved Version
Installation Type	Specifies installation requirement	New Installation / Update
Business Justification	Explains the business need	Project or job requirement
Required By Date	Captures requested completion date	Selected Date
Manager / Approver	Identifies approval authority	Selected Approver
RITM	Tracks the submitted catalog request	Auto-generated
Catalog Task	Tracks IT fulfillment	Auto-generated task

Figure 2 : Service Catalog Item and Related Request Records Catalog Item → Requested Item (RITM) → Approval → Catalog Task → Closure

5. LICENSED SOFTWARE REQUEST FORM CONFIGURATION

The Licensed Software Installation Request catalog item is the central employee-facing form. It captures the information required to make an approval and fulfillment decision while reducing incomplete or unclear software requests.

5.1 Catalog Item Creation

- Open Service Catalog and create a new catalog item.
- Enter the item name as Licensed Software Installation Request.
- Add a suitable short description and detailed description.
- Select the appropriate catalog and category.
- Configure the item variables required for software installation.
- Mark important fields such as software name, business justification and requested-for user as mandatory.
- Configure the item to be visible to the intended employee users.
- Save the item and test it from the employee Service Catalog view.

5.2 Request Classification

FIELD / VARIABLE	VALUE / CONFIGURATION
Requested For	Selected User / Current User
Software Name	Selected approved software
Software Version	Selected or entered approved version
Installation Type	New Installation / Upgrade / Reinstallation
Business Justification	Mandatory text explaining the requirement
Required By Date	Employee-selected date
Additional Comments	Optional installation information
Approval Status	Pending / Approved / Rejected
Fulfillment Status	Open / In Progress / Completed / Closed

After entering the required information, the employee submits the catalog item. ServiceNow generates the corresponding Requested Item and starts the configured approval and fulfillment process.

Figure 3 : Licensed Software Installation Request Form [Employee Information] [Software Details] [Business Justification] [Required Date] [Submit]

6. APPROVAL AND FULFILLMENT AUTOMATION

Approval and fulfillment automation ensures that licensed software is not installed solely from an unverified employee request. The workflow/Flow Designer process evaluates the request, obtains approval from the configured approver, and creates or routes fulfillment work to the appropriate IT team.

6.1 Approval Validation

- Submit a Licensed Software Installation Request with all mandatory information.
- Verify that a Requested Item is generated successfully.
- Verify that the request enters the configured approval stage.
- Open the approval record using the assigned approver.
- Approve the request and verify that fulfillment processing continues.
- Reject a test request and verify that the request does not proceed to installation.
- Verify that the requester receives the configured status notification.
- Verify that an approved request creates or assigns the required Catalog Task.

6.2 Benefits

- Provides controlled approval before licensed software installation.
- Improves software license governance and traceability.
- Reduces manual email communication between employees, managers and IT.
- Ensures IT receives complete and standardized request information.
- Provides clear request status visibility to the requester.
- Creates a foundation for license inventory integration, notifications and reporting.

6.3 Suggested Flow

Figure 4 : Approval and Fulfillment Flow Submit Request → Validate → Manager/License Approval → Approved? → Yes: Create Fulfillment Task → IT Installs Software → Complete No: Notify Requester → Close / Cancel

7. DATA VALIDATION AND PROCESS CONTROL

Data validation and process control ensure that software requests contain reliable information and follow the defined approval and fulfillment rules. Mandatory variables, reference fields, controlled choices and flow conditions help maintain consistent request records.

7.1 Default Values Configuration

- Configure Requested For to default to the current logged-in user where appropriate.
- Configure default request status according to the catalog process.
- Provide controlled values for Installation Type.
- Use reference or choice values for software selection where applicable.
- Verify that default values appear correctly when a new request is opened.

7.2 Validation Rules

- Verify that mandatory request information is entered before submission.
- Validate the software selection and requested version.
- Validate the business justification field.
- Validate that the required-by date is appropriate for the request.
- Verify that the approval flow receives the required information.
- Prevent incomplete requests from progressing to fulfillment.

7.3 Error Handling and User Feedback

- Attempt to submit a request with missing mandatory information.
- Verify that the catalog form identifies the missing information.
- Submit a request with valid information and verify successful creation.
- Reject an approval test case and verify that the requester is informed.
- Verify that fulfillment failures or incomplete information can be returned for correction.
- Confirm that request status remains traceable throughout the process.

This stage demonstrates that the Service Catalog process controls the quality of request information while providing clear feedback to employees and support teams.

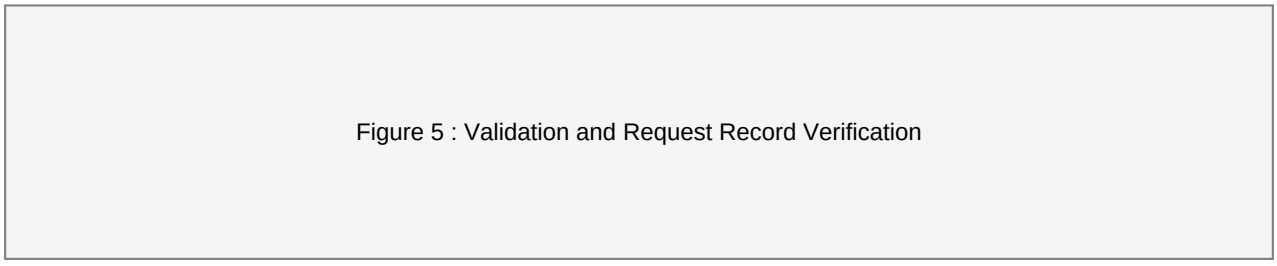

Figure 5 : Validation and Request Record Verification

8. UPDATE SET MANAGEMENT

8.1 Update Set Creation

An Update Set is used to capture the ServiceNow configuration developed for the Licensed Software Installation Request project so that the configuration can be reviewed and moved between instances as required.

- Create an Update Set for the Licensed Software Installation Request project.
- Set the Update Set as Current during configuration work.
- Configure the catalog item, variables, related flows and supporting configuration.
- Verify that the required configuration changes are captured.
- Review the Update Set before completion.

8.2 Configuration Capture Verification

- Open the project Update Set.
- Review the Customer Updates captured in the Update Set.
- Verify that the catalog item configuration is included where applicable.
- Verify that related configuration components are captured.
- Confirm that the Update Set can be completed and prepared for deployment.

Update Set management provides controlled configuration movement and supports maintainability of the ServiceNow implementation.

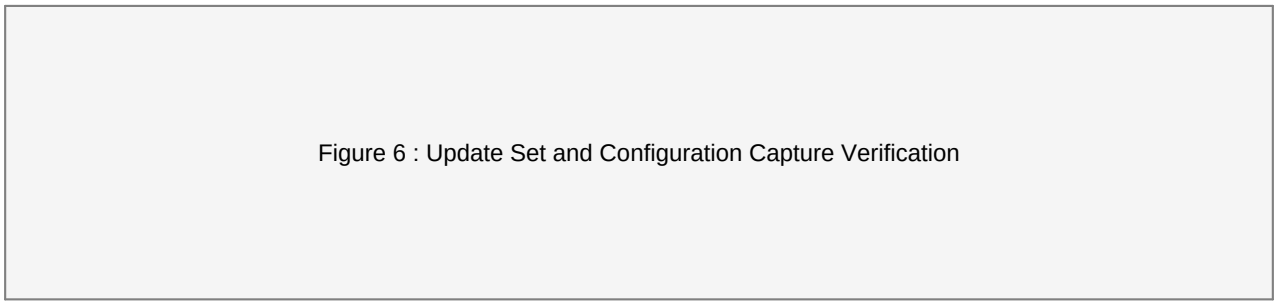

Figure 6 : Update Set and Configuration Capture Verification

9. TESTING AND FUTURE ENHANCEMENTS

9.1 Functional Testing

TEST	EXPECTED RESULT
Open Catalog Item	Licensed Software Installation Request is visible
Enter Mandatory Fields	Valid values are accepted
Submit Request	Requested Item is created successfully
Approval	Request is routed to configured approver
Reject Request	Request does not proceed to fulfillment
Approve Request	Fulfillment processing starts
Catalog Task	IT fulfillment task is created/assigned
Complete Installation	Task and request status are updated
Notification	Requester receives configured status update
Update Set	Required configuration is captured

- Verify that all configured functions work as expected.
- Confirm that valid requests are created successfully.
- Confirm that invalid or incomplete requests are controlled.
- Confirm that approval and fulfillment transitions occur correctly.

9.2 Future Workflow Automation

- Integrate with a software/license inventory or asset management process.
- Automatically check license availability before approval or fulfillment.
- Configure multi-level approval for high-cost or restricted software.
- Send automated notifications for pending approvals and overdue fulfillment.
- Add SLA monitoring for software installation requests.

9.3 Reporting and Dashboard Opportunities

- Number of software installation requests by month.
- Requests by software application and version.
- Approved, rejected, pending and completed requests.
- Average approval and fulfillment time.
- Requests by department or employee group.
- Overdue requests and fulfillment SLA performance.

10. FINAL VALIDATION, TESTING AND CONCLUSION

10.1 Final Validation

- Licensed Software Installation Request catalog item is created and accessible.
- Required catalog variables are available and functioning.
- Mandatory field and validation rules work correctly.
- Requested Item records are generated successfully.
- Approval processing works according to the configured flow.
- Approved requests are routed to the IT fulfillment team.
- Rejected requests are prevented from continuing to installation.
- Request status and fulfillment information are maintained correctly.
- Required configuration is captured in the project Update Set.

10.2 Testing and Deployment Validation

- End-to-end software request submission.
- Catalog variable and field validation.
- Approval and rejection scenarios.
- Fulfillment task creation and assignment.
- Status update and notification verification.
- Request closure after successful installation.
- Update Set configuration capture.
- Accuracy and completeness of request information.
- Readiness for future license and asset management integration.

10.3 Project Outcome

The project demonstrates how ServiceNow Service Catalog can be used to standardize the employee software installation request process. The solution connects the employee-facing catalog item with structured request variables, approval processing, IT fulfillment tasks and request tracking. This creates a controlled and traceable process for licensed software installation.

10.4 Conclusion

The Licensed Software Installation Request Management System in ServiceNow provides a structured approach to managing software requests from employee submission through approval, fulfillment and closure. The implementation focuses on Service Catalog configuration, request data collection, validation, approval automation, fulfillment task management and Update Set control.

The solution reduces manual request handling, improves visibility, supports software governance and provides a strong foundation for future enhancements such as license availability checking, SLA management, asset integration, automated notifications and management dashboards.

APPENDIX: PROJECT PHASE SUMMARY

PHASE	MAIN ACTIVITY	EXPECTED RESULT
1 Requirement Analysis & Planning	Define objectives, scope, stakeholders and roadmap	Requirements and implementation plan defined
2 Service Catalog Configuration	Create category and Licensed Software Installation catalog item	Employee-facing catalog request is available
3 Form & Data Configuration	Configure variables, mandatory fields and validation	Standardized software request information
4 Approval & Fulfillment Automation	Configure approval and IT fulfillment process	Controlled request processing from approval to installation
5 Testing, Validation & Update Set Management	Test request scenarios and verify configuration capture	Validated process and deployment-ready configuration
6 Future Automation	Plan license checks, SLA, notifications, reports and integrations	Scalable foundation for advanced automation
7 Conclusion	Document project outcome	Structured licensed software request solution

Project Title: An employee requests installation of licensed software through the Service Catalog

Platform: ServiceNow

Primary Service: Licensed Software Installation Request

Core ServiceNow Components: Service Catalog, Catalog Variables, Requested Item, Approval, Flow Designer, Catalog Task, Notifications and Update Sets

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